

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT202		INTELLIGENT SCIENCE	

2nd SEMESTER 2022/23 RESIT EXAMINATION

UNDERGRADUATE – YEAR 3

COMPLEXITY OF ALGORITHMS

TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

- 1. This is a closed-book examination.**
- 2. The examination will be conducted in all other respects in accordance with the Regulations for the Conduct of Examinations.**
- 3. Total marks available are 100. This exam will count for 100% of the final mark.**
- 4. Answer all questions. There is NO penalty for providing a wrong answer.**
- 5. Only English solutions are accepted.**

Question 1. [5 MARKS]

Suppose that we have numbers between 1 and 1000 in a binary search tree, and we want to search for the number 363. Which of the following sequences could not be the sequence of nodes examined? Please explain in detail.

- a. 935, 278, 347, 621, 299, 392, 358, 363.
- b. 924, 220, 911, 244, 898, 258, 362, 363.
- d. 2, 399, 387, 219, 266, 382, 381, 278, 363.

[5 marks]

Question 2. [5 MARKS]

Illustrate the result of each operation in the sequence ENQUEUE(Q, 4), ENQUEUE(Q, 1), ENQUEUE(Q, 3), DEQUEUE(Q), and ENQUEUE(Q, 8) on an initially empty queue Q.

[5 marks]

Question 3. [10 MARKS]

Use the substitution method to prove that the recurrence

$$T(n) = T(n-1) + \Theta(n)$$

has the solution

$$T(n) = \Theta(n^2)$$

[10 marks]

Question 4. [10 MARKS]

Show the process of insertion of element [10] to the max heap A[15, 13, 9, 5, 12, 8, 7, 4, 0, 6, 2, 1]. Please show original heap, all intermediate steps and final heap.

[10 marks]

Question 5. [10 MARKS]

Can the master method be applied to the recurrence

$$T(n) = 4T(n/2) + n^2 \lg n$$

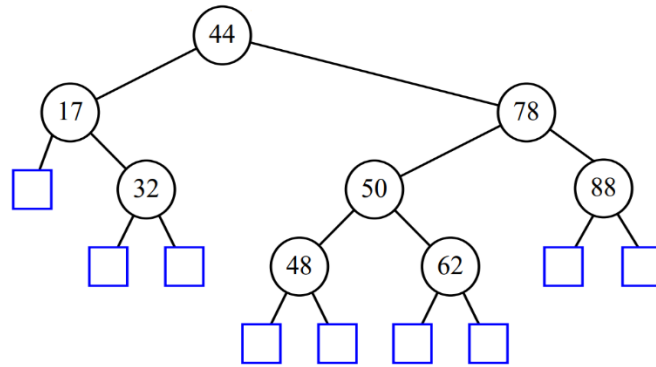
Why or why not? Give an asymptotic upper bound for this recurrence.

[10 marks]

Question 6. [10 MARKS]

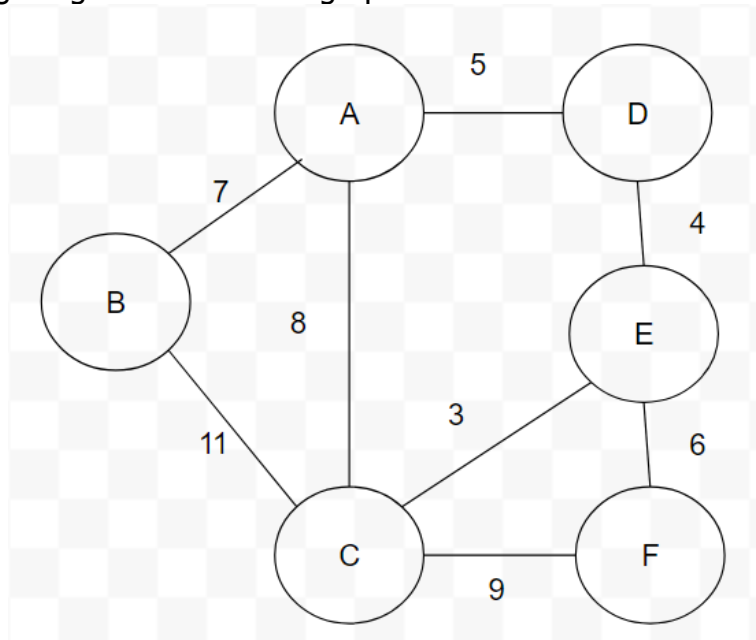
Draw the AVL tree resulting from the deletion of the item with key 32 from the following AVL tree. Please also show all intermediate steps.

[10 marks]



Question 7. [7+3=10 MARKS]

Given the following weighted undirected graph:



- (1) Show the step by step process of using the Prim's algorithm to find the minimum spanning tree. We start with node A. [7 marks]
- (2) What is the total weight of the minimum spanning tree? [3 marks]

Question 8. [7+8+5=20 MARKS]

Alice and Bob wish to use the RSA cryptosystem for communication. Alice get a pair of prime numbers 13, 17.

- (1) What is the public modulus n , and $\phi(n)$? [7 marks]
- (2) Alice choose a public key 77, what is the corresponding private key? [8 marks]
- (3) Encrypt message 2 with the public key 77. [5 marks]

Question 9. [5+10+5=20 MARKS]

An Independent Set S of graph $G = (V, E)$ is a set of vertices such that no two vertices in S are adjacent to each other. It consists of non-adjacent vertices. The Independent Set problem is to determine if the graph contains an independent set of vertices of at least size k .

- (1) Show that the Independent Set problem is in NP. [5 marks]
- (2) Derive a reduction from 3SAT to the Independent Set problem, and justify it. [10 marks]
- (3) Derive a reduction from the Independent Set problem to the Vertex Cover problem, and justify it. [5 marks]

END OF EXAM PAPER